



In situ growth of nickel cobalt sulfide on carbon cloth by electrodeposition as binder-free electrode for supercapacitor

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ABSTRACT

Development of binder-free electrode of pseudocapacitive material with excellent cycling stability for supercapacitor shows great attraction in energy storage market. Here, nickel cobalt sulfide (NCS) has been successfully grown on carbon cloth (CC) by electrodeposition in a sulfate-based electrolyte solution with thiourea (TU) as sulfur source and chelation agent. The results show that the chemical composition, microstructure and electrochemical performance of the as-prepared electrode can be modulated by tuning the deposition overpotential. The oriented growth of NCS nanosheets driven by electric field coupled with thiourea adsorption endows the deposits with intersected porous nanosheet-array microstructures, which enables the fast electronic and ionic transmissions. The optimal electrode prepared under -1.0 V (vs Ag/AgCl) (denoted as NC2S200-1.0/CC) can deliver a discharge specific capacitance of 909.5 F g^{-1} at 1 A g^{-1} , which can maintain 89.2% at 10 A g^{-1} , demonstrating outstanding rate performance. As successively charging and discharging for 4000 cycles at 5 A g^{-1} , 120.6% of the initial value can be retained, illustrating exceptional cyclic stability. In addition, the assembled NC2S200-1.0/CC//AC asymmetric supercapacitor can exhibit an energy density of 26.2 Wh kg^{-1} which corresponds to a power density of 821.1 W kg^{-1} , when charging-discharging at 2 A g^{-1} for 10000 cycles, a capacitance retention of 75.5% can be achieved, manifesting good cycling stability. This work provides a new idea for fabricating transition metal sulfides supported binder-free electrode by electrodeposition.

1. Introduction

The development of science of energy conversion and storage has stimulated considerable researchers to exploit advanced materials for supercapacitors on account of the attractive features of them, such as extremely high power densities and ultralong lifetimes [1,2]. Compared with other constituents of the device, the electrode materials play a significant role in the whole performance of a supercapacitor. Specifically, the active surfaces, the dimensions of primary particles, the crystallinities, the porosities and voids, the electronic conductivities, the mechanical strengths and structural stabilities of the electrode materials immediately influence the capacitances of the electrodes and the kinetics of electrode processes [3–5]. Although a great deal of efforts has been devoted to the design and synthesis of advanced electrode materials for supercapacitors, many challenges still exist in terms of process

simplification, energy saving, low contamination and sustainability, and so on. Therefore, a large number of sustained works would be expected to advance the material research for the commercialization of supercapacitors.

Among a wide variety of materials for supercapacitors, bimetallic sulfides, such as NiCo_2S_4 [6–8], MnCo_2S_4 , ZnCo_2S_4 , FeCo_2S_4 [11,12] and CuCo_2S_4 [13,14], have evoked worldwide interests in the past decades owing to their high conductivities and high capacitances [15], which is of vital importance for the supercapacitor applications. Compared with other sulfides, NiCo_2S_4 has attracted more attention due to its well-established synthetic route and well-defined energy storage mechanism. Up to now, NCSs with a variety of morphologies including mesoporous nanoparticles [16,17], urchin-like nanocrystals [18], ball-in-ball hollow spheres [19], cockscomb flower-like nanostructures [20], nanosheet arrays [21,22], ultrathin nano-petals [23] and 3-D

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honeycombs [24] have been successfully prepared via different synthetic approaches. With regard to the preparation process, electrodeposition exhibits intriguing advantages in light of its simplification, controllability, cost-effective, environmental friendliness and safety [22, 25–27]. Wang et al. fabricated a NCS/RGO/NF free-standing electrode through a cyclic voltammetry deposition in an aqueous solution containing CoCl_2 , NiCl_2 and TU, which exhibited outstanding capacitive performance [28]. Li et al. successfully achieved a NCS/polypyrrole/carbon paper(NCS/PPy/CP) electrode by an unipolar pulse electrodeposition in an electrolyte solution containing $\text{Ni}(\text{NO}_3)_2$ and $\text{Co}(\text{NO}_3)_2$ with sublimed sulfur as sulfur source [29]. The results indicated that the optimal molar ratio (MR) of Ni to Co was 1:1 and the corresponding electrode delivered a capacitance of 2004 F g^{-1} at 1 A g^{-1} . Dong et al. constructed $\text{NiCo}_2\text{S}_4/\text{CP}$ electrode by a potentiostatic electrodeposition in an electrolyte solution containing $\text{Ni}(\text{NO}_3)_2$, $\text{Co}(\text{NO}_3)_2$ and TU [30]. Chen et al. applied a cyclic voltammetry deposition to construct NCS nanosheet arrays on CC in an electrolyte solution containing CoCl_2 , NiCl_2 and TU and outlined the influence of the molar ratio of Ni to Co on the electrochemical performance of the as-prepared binder-free electrodes [31]. As known, the electro-crystallization of a material is greatly influenced by the deposition modes, deposition parameters, electrolyte solution components and deposition substrates [32–34]. Moreover, for the electrodeposition of NCS, the electrolyte composition plays a vital role in the deposition reaction of NCS and which also influences the elemental composition and microstructure of the deposit. Despite of the above reported works, to the best of our knowledge, the electrodeposition of NCS on flexible CC in an electrolyte solution of nickel cobalt sulfates with TU as sulfur source has not been reported yet. Therefore, it is necessary to explore the deposition mechanism of NCS in the sulfate-based electrolyte solution and investigate the charge storage properties of the sediments.

Bear this in mind, herein, a series of NCSs, denoted as $\text{NiCo}_2\text{S}_{200-x}$, have been in situ grown on flexible CC as free-standing electrodes for supercapacitors by a potentiostatic deposition mode in an electrolyte bath containing nickel sulfate, cobalt sulfate and TU. By tuning the deposition potential, the chemical compositions and morphologies of the deposits can be easily regulated, leading to different capacitive behaviors of the as-prepared free-standing electrodes. Among these electrodes, the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode that obtained under -1.0 V vs Ag/AgCl exhibited excellent rate and cyclic performance. Moreover, the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}/\text{AC}$ HC delivered an energy density of 26.2 Wh kg^{-1} at 821.1 W kg^{-1} and showed good cycling performance of 75.5 % capacitance retention at 2 A g^{-1} for 10000 cycles.

2. Experimental

2.1. Materials

All the chemicals used here are all analytical reagents and used as received. All the electrolyte solutions are prepared using de-ionized water. The CC is washed with ethanol and de-ionized water for 15 min in an ultrasonic cleaner, respectively.

2.2. Electrodes preparation

The $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ electrodes were prepared by a one-step potentiostatic electrodeposition that was performed in a three-electrode configuration with CC, Pt plate($1 \text{ cm} \times 1 \text{ cm}$) and Ag/AgCl ($3 \text{ mol L}^{-1} \text{ KCl}$) electrode as the working electrode (WE), counter electrode (CE) and reference electrode (RE), respectively. The electrolyte solutions were prepared by dissolving $1.31 \text{ g NiSO}_4 \cdot 6\text{H}_2\text{O}$, $2.81 \text{ g CoSO}_4 \cdot 7\text{H}_2\text{O}$ and $76.12 \text{ g CH}_4\text{N}_2\text{S}$ in 150 mL de-ionized water and then diluted to 1 L . All the electrodepositions were carried out under different potentials from -1.2 to -0.8 V (vs. Ag/AgCl) for 15 min under room temperature on a CHI660A electrochemical workstation. For convenience, the as-prepared electrodes were denoted as $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ (x

$= 0.8, 0.9, 1.0, 1.1$ and 1.2) where $x = 0.8, 0.9, 1.0, 1.1$ and 1.2 refer to the deposition potentials of $-0.8, -0.9, -1.0, -1.1$ and -1.2 V , respectively. Then, the deposits were cleaned and dried before the weighing and the mass loadings of the deposits were $0.62, 0.53, 0.70, 0.74$ and 0.70 mg cm^{-2} , respectively, for the $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ electrodes when $x = 0.8, 0.9, 1.0, 1.1$ and 1.2 .

2.3. Fabrication of hybrid supercapacitor

The hybrid supercapacitor (HC) $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}/\text{AC}$ was assembled into a coin cell using the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode and the activated carbon (AC) electrode as the cathode and anode, respectively, with filter paper as the separator and $2 \text{ mol L}^{-1} \text{ KOH}$ as the electrolyte solution. The AC electrode was prepared according to our previous work [35]. The active mass ratio of cathode to anode was calculated based on the charge-balance equation (1) as follows:

$$\frac{m_+}{m_-} = \frac{C_- V_-}{C_+ V_+} \quad (1)$$

Where m_+ and m_- , C_+ and C_- , V_+ and V_- represent the mass loadings, discharge specific capacitances (DSCs), discharge potential windows of cathode and anode, respectively.

2.4. Material characterizations

The phase of the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode was characterized by X-ray diffraction (XRD, Rint-2000, Rigaku). The functional group of $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode was characterized by Raman spectra (Renishaw, inVia, 532 nm). The micromorphologies of the $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ electrodes were characterized by scanning electron microscopy (SEM, Sirion 200) and transmission electron microscopy (TEM, Tecnai G12). The elemental compositions and distributions of the $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ electrodes were characterized by energy dispersive X-ray spectroscopy (EDS). The chemical composition and elemental chemical states of the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode were characterized by X-ray photoelectron spectroscopy (XPS, Thermal fisher Escalab Xi+). The Brunauer-Emmett-Teller (BET) surface area and pore distribution of the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode were measured by a JW-BK200A analyzer.

2.5. Electrochemical measurements

The galvanostatic charge-discharge (GCD) tests were carried out on a Land battery tester using the $\text{NiCo}_2\text{S}_{200-x}/\text{CC}$ electrodes, Pt, Hg/HgO electrode and $2 \text{ mol L}^{-1} \text{ KOH}$ as the WEs, CE, RE and electrolyte solution, respectively. The electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) were measured on a CHI660A electrochemical workstation. The DSCs of the electrodes and HC were computed by integrating the discharge curves according to equation (2) [25]:

$$C_m = \frac{2I \int_{t_i}^{t_f} V dt}{m \Delta V^2} \quad (2)$$

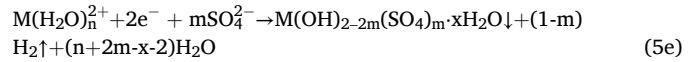
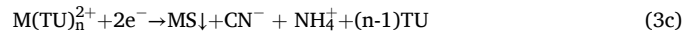
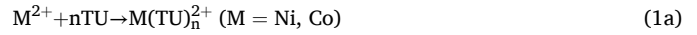
In this equation, C_m indicates the DSC (F g^{-1}), I means the current (A), t_i and t_f represent the initial and final discharge times (s), respectively, V and ΔV mean the potential (V) and discharge potential window (V), respectively, m denotes the mass of the active material (g). But, for the HC, m denotes the total mass of active materials of the cathode and anode. Additionally, the energy density (Wh kg^{-1}) and power density (W kg^{-1}) of the HC were computed according to our publications [27, 35].

3. Results and discussions

3.1. Electrodeposition mechanism of the NCS/CC electrode

Fig. 1 shows the preparation process of the NCS/CC electrode and the electrodeposition mechanism of NCS under electric field. As seen in Fig. 1a—a core-shell structure free-standing electrode with carbon fiber (CF) and NCS as the core and shell layers can be easily prepared by a potentiostatic deposition in a three-electrode configuration with CC, Pt and Ag/AgCl electrode as the WE, CE and RE, respectively. The corresponding electrodeposition mechanism is vividly displayed in Fig. 1b. In detail, when an external electric field is imposed, the metal ion complexes $M(TU)_n^{2+}$ ($M = Ni, Co$) that formed by reaction (1a) move automatically from the bulk solution to the CF surface under the electric field force, while SO_4^{2-} ions move oppositely to the anode. During this process, due to the steric hindrance and electrostatic attraction of SO_4^{2-} ions, the electrical migration rates of the $M(TU)_n^{2+}$ ions may be retarded, which may influence the morphologies of the deposits. On the substrate surface, TU molecules and $M(TU)_n^{2+}$ ions are vertically adsorbed [36,37] and then can accept the injected electrons, leading to the electro-reduction of TU and $M(TU)_n^{2+}$ to S^{2-} ions, as described by reactions (2b) and (3c), that can react with M^{2+} to produce $NiCo_2S_4$ precipitates, our target product, which is illustrated by reaction (4d). However, some metal hydroxides intercalated by H_2O molecules and SO_4^{2-} anions ($M(OH)_{2-2m}(SO_4)_m \cdot xH_2O$) may be formed simultaneously during the

electrodeposition of $NiCo_2S_4$ owing to the side effect as shown as reaction (5e). Thus, the elemental compositions of the resultant deposits can be different under various deposition potentials, which can be proved by the following EDS measurements [25,27,38,38].



3.2. Materials characterizations

Fig. 2a shows the XRD pattern of the $NiCo_2S_4$ /CC electrode. It is clear that there is no evident characteristic diffraction peaks associated with $NiCo_2S_4$ can be observed except for the peaks of CC substrate, indicating the low crystallinity of the deposited NCS. The result is analogous to those reported by us and other groups regarding to the electrodeposition of transition metal hydroxides and sulfides [25,27]. In order to identify the structural information of the $NiCo_2S_4$ /CC electrode, its Raman spectra is also provided in Fig. 2b. As seen, the

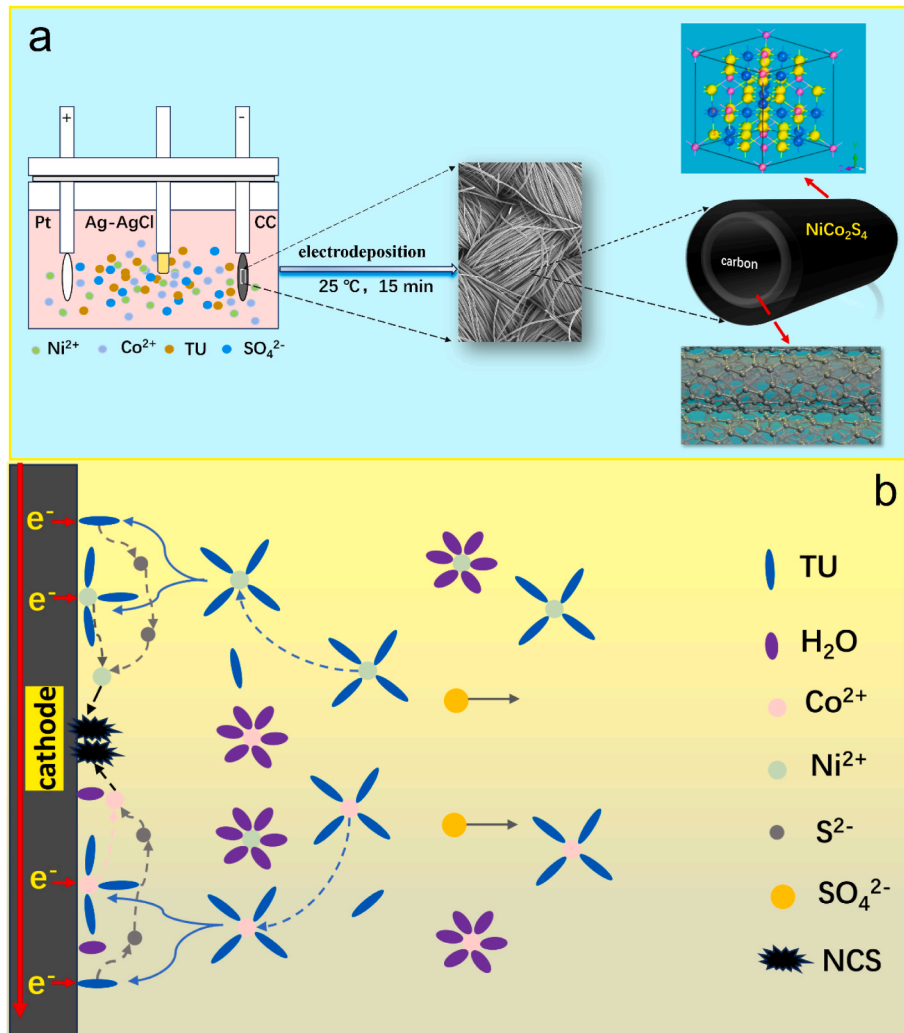


Fig. 1. (a) Schematic illustration of the preparation of NCS/CC electrode, (b) Diagram of the deposition mechanism of NCS under electric field.

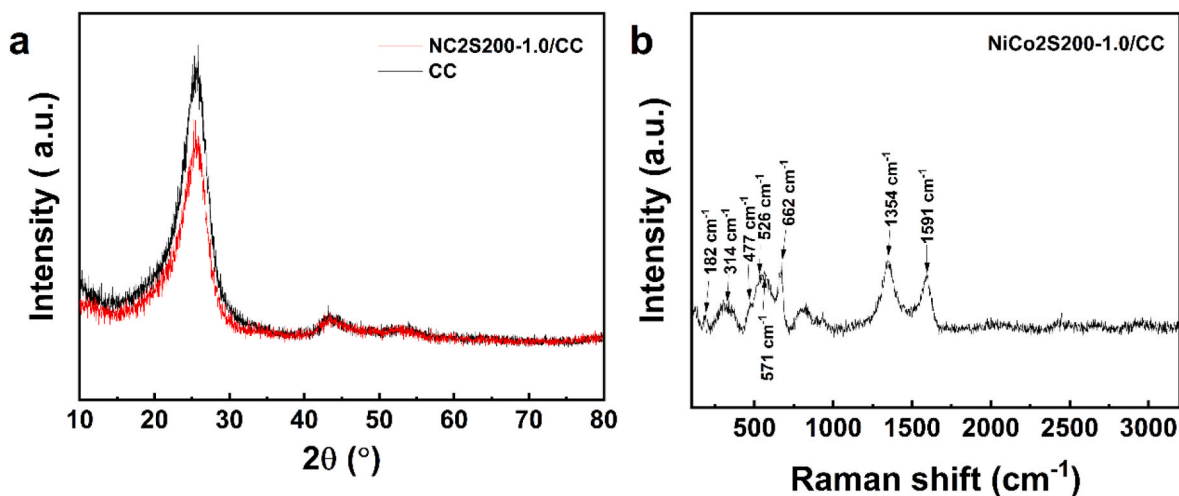


Fig. 2. (a) XRD pattern and (b) Raman spectra of the NiCo2S200-1.0/CC electrode, respectively.

Raman peaks located at 182, 314, 477, 526, 571 and 662 cm^{-1} can be assigned to the F_{2g} , F_{2g} , E_{2g} , T_{2g} , A_{1g} and A_{1g} vibrational modes of NiCo_2S_4 [7,30,39]. Besides, the peaks at 1354 and 1591 cm^{-1} belong to the characteristic D (disordered) band and G (graphitic) band of CC [7,40–42]. Therefore, it can be concluded that the active material of the

$\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode should predominately be NiCo_2S_4 . To disclose the elemental composition and chemical states of the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode, the XPS spectra are provided in Fig. 3. From the survey spectra (Fig. 3a), the signals of Ni, Co, S, O and C can be detected, indicating the coexistence of these elements. Here, the

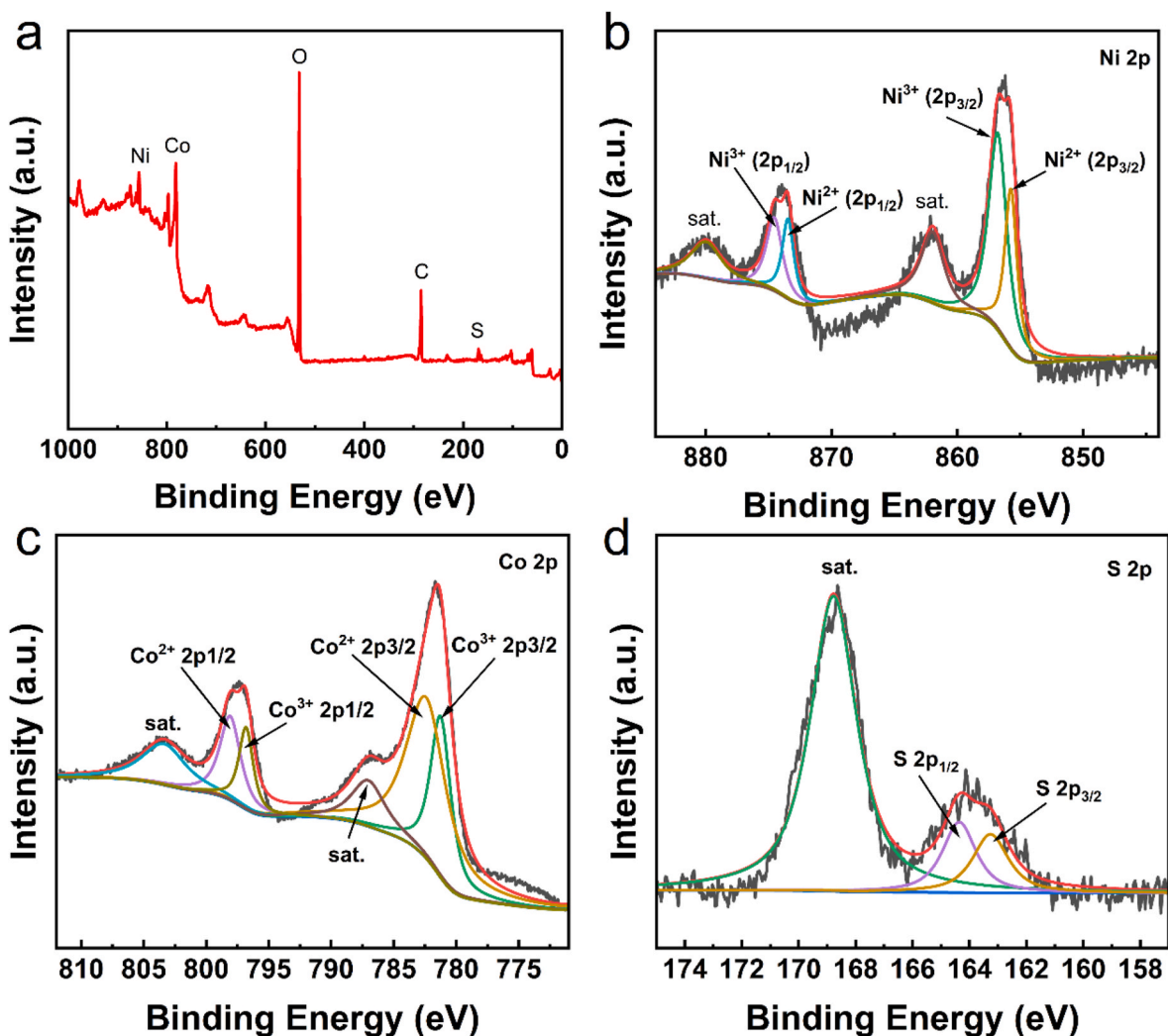


Fig. 3. (a) XPS survey spectra of the $\text{NiCo}_2\text{S}_{200-1.0}/\text{CC}$ electrode, high resolution XPS spectra of Ni 2p (b), Co 2p(c) and S 2p(d).

appearance of O element should be related to the byproduct of $M(OH)_{2-2m}(SO_4)_m \cdot xH_2O$, while C comes mainly from CC, these observations are identical with the EDS analysis. Furthermore, based on Gaussian optimization, the high-resolution spectra of Ni 2p (Fig. 3b) can be fitted into two couples of spin-orbit doublet peaks centered at 873.4 and

855.8 eV, 874.5 and 856.9 eV which correspond to Ni 2p_{1/2} and Ni 2p_{3/2} of Ni²⁺ and Ni³⁺, respectively. While the doublet peaks centered at 880.1 and 862.0 eV correspond to the satellite peaks of Ni 2p [17,22]. For the Co 2p (Fig. 3c), the fitted peaks at 796.7 and 781.3 eV correspond to Co 2p_{1/2} and Co 2p_{3/2} of Co³⁺, while those located at 798.1 and

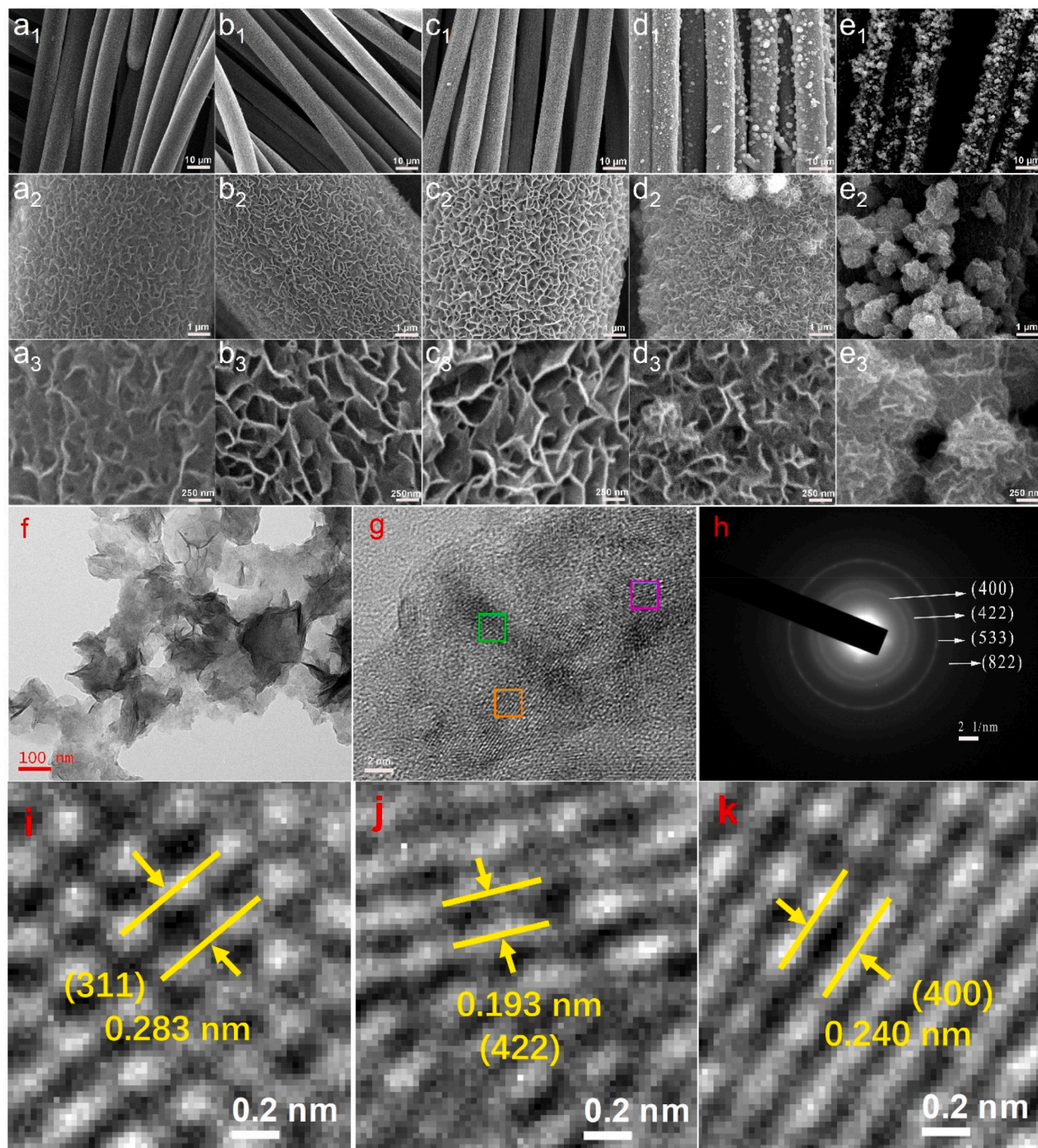


Fig. 4. SEM images of the (a₁-a₃) NiCo₂S₂₀₀-0.8/CC, (b₁-b₃) NiCo₂S₂₀₀-0.9/CC, (c₁-c₃) NiCo₂S₂₀₀-1.0/CC, (d₁-d₃) NiCo₂S₂₀₀-1.1/CC and (e₁-e₃) NiCo₂S₂₀₀-1.2/CC electrodes, respectively. (f) TEM image, (g) HRTEM, (h) SAED images of the NiCo₂S₂₀₀-1.0/CC electrode. Inverse FFT images of the areas marked by the green (i), pink (j) and orange (k) square frames in Fig. 3(g). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

782.6 eV belong to Co^{2+} , with two satellite peaks located at 803.5 and 787.1 eV [43,45]. The S 2p (Fig. 3d) can also be fitted into two peaks at 164.4 and 163.3 eV which correspond to S $2p_{1/2}$ and S $2p_{3/2}$ of S [2–22, 44]. These results demonstrate the chemical valences of Ni and Co are Ni (II), Ni(III), Co(II) and Co(III), well accordance with the other reported works [17].

In order to consolidate our above conclusion, TEM, HRTEM and SAED images of the active material of the NiCo₂S₂₀₀–1.0/CC electrode are also displayed in Fig. 4f–h. As shown in Fig. 4f, some slightly stacked nanoflakes can be clearly observed. Moreover, the corresponding HRTEM image (Fig. 4g) and inverse FFT images (Fig. 4i–k) exhibit characteristic interlayer spacings of 0.283, 0.240 and 0.193 nm that indexed to the (311), (400) and (422) lattice planes of NiCo₂S₄ (PDF# 20–0782). In addition, the SAED image (Fig. 4h) shows four diffraction rings that correspond to the (400), (422), (533) and (822) lattice planes of NiCo₂S₄, further verifying the existence of NiCo₂S₄ in the deposits of the NiCo₂S₂₀₀–1.0/CC electrode. Besides, the elemental mapping graphic of the NiCo₂S₂₀₀–1.0/CC electrode is also supplemented in Fig. S1. It is clear that Ni, Co and S precipitate homogeneously on the surface of CC, authenticating the successful electrodeposition of NiCo₂S₄. In view of the importances of specific surface and pore structure on the capacitive performance of electrode material, the BET specific surface area (SSA) and pore distribution of the NiCo₂S₂₀₀–1.0/CC electrode are also measured by N₂ adsorption-desorption isotherms and the results are supplied in Fig. S2.

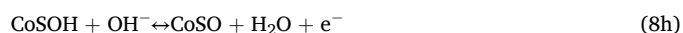
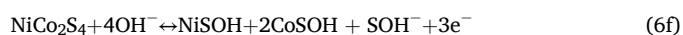
As seen in Fig. S2a, evident IV type isotherm can be observed in the pressure range of $0.2 < p/p_0 < 1.0$, indicating the presence of mesopores in the deposits [45]. However, the narrow hysteresis loop also demonstrates the small SSA of the electrode which is only $4.6 \text{ m}^2 \text{ g}^{-1}$. The tiny value of SSA should be ascribed to the low loading of NCS (0.7 mg cm^{-2}). Moreover, the most probable aperture of the deposit is calculated to be 4.4 nm based on BJH method, further verifying the pores in the NCS are predominately mesopores, which is helpful for the capacitance enhancement of the electrode [45].

For the purpose of investigating the influence of deposition potential on the micromorphologies of the deposited NCSs, SEM images of the NiCo₂S₂₀₀-x/CC electrodes are presented in Fig. 4. As seen, all the CFs are covered by NCSs of different morphologies. Detailly, for the NiCo₂S₂₀₀–0.8/CC electrode (Fig. 4(a₁–a₃)), some intertwined nanosheets with obscure nanopores can be observed, which can be favorable for the permeation of electrolyte solution. However, in comparison, the NiCo₂S₂₀₀–0.9/CC (Fig. 4(b₁–b₃)) and NiCo₂S₂₀₀–1.0/CC electrodes (Fig. 4(c₁–c₃)) exhibit dense nanosheet-arrays with well-defined nanoporous architectures. While, a number of microbuds emerge on the surface of nanosheets for the NiCo₂S₂₀₀–1.1/CC electrode (Fig. 4(d₁–d₃)). What's more, a great number of flower-like clusters of particles can be observed for the NiCo₂S₂₀₀–1.2/CC electrode (Fig. 4(e₁–e₃)). These phenomena demonstrate that the deposition rate and the amount of NCSs increase gradually with increasing the deposition overpotential. Moreover, when switching the potential from -0.8 V to -1.0 V , especially for -0.9 V and -1.0 V , slightly changes can be observed, but, when lowering down the potential to -1.1 and -1.2 V , apparent changes occur, which suggesting that the modulation of deposition potential plays a vital role in controlling the morphologies of deposits. On another aspect, the influence of deposition potentials on the elemental compositions are also investigated and the results are presented in Fig. S3. From the EDS results, the variations of the contents of Ni, Co, S and O can be explicitly observed. In detail, the MR of Co to Ni changes in a quasi-parabola style, namely the MR increase firstly and then decrease with the increment of deposition overpotential. Surprisingly, the MR turns to be approximately 2:1 for the NiCo₂S₂₀₀/CC electrode, which is very close to the stoichiometric coefficient of NiCo₂S₄. Additionally, the MRs of S to O are 1.47, 7.65, 3.32, 2.93 and 2.88 for the NiCo₂S₂₀₀-x/CC electrodes when $x = 0.8, 0.9, 1.0, 1.1$ and 1.2 , respectively. Remarkably, the vulcanization degrees of the deposits are greatly influenced by the deposition overpotential. However, the underlying

mechanism is still unknown and needs to be further studied. All in all, the NiCo₂S₂₀₀–1.0/CC and NiCo₂S₂₀₀–0.9/CC electrodes exhibit analogous micromorphologies and elemental constituents.

3.3. Electrochemical evaluations

The electrochemical behaviors of the NiCo₂S₂₀₀-x/CC electrodes are systematically investigated and the results are exhibited in Fig. 5. Firstly, cyclic voltammetric measurements are employed to explore the charge storage mechanisms of the obtained electrodes. As shown in Fig. 5a, all the CV curves which collected in a potential window of 0 – 0.6 V (vs. Hg/HgO) present obvious redox peaks, illustrating their pseudo-capacitive charge storage mechanisms [46]. However, some distinctive differences in the CV contours still can be observed, especially for the peak potentials and the intensities of peaks as well as the areas of CV loops. Among these curves, the NiCo₂S₂₀₀–0.8/CC, NiCo₂S₂₀₀–0.9/CC and NiCo₂S₂₀₀–1.0/CC electrodes present two pairs of broad redox peaks with weak peak currents, showing evident multi-electron transfer reaction characteristics, which are associated with the redox reactions between Ni^{2+} and Ni^{3+} , Co^{2+} and Co^{3+} as well as Co^{3+} and Co^{4+} . The corresponding reactions can be expressed as reactions (6f)–(8h), as below [4,4,4,6,6].



While, only one couple of broad redox peaks can be seen for the NiCo₂S₂₀₀–1.1/CC and NiCo₂S₂₀₀–1.2/CC electrodes, suggesting the merges of redox peaks in these electrodes, however, the underlying reaction mechanisms of them are identical with the other electrodes. Combined with the EDS results, it can be deduced that the variations of peak potentials should be related to the elemental composition of the deposits. Besides, from the areas of the CV loops, it can be speculated that the DSCs of these electrodes should follow a rank of NiCo₂S₂₀₀–1.0/CC < NiCo₂S₂₀₀–0.9/CC < NiCo₂S₂₀₀–0.8/CC < NiCo₂S₂₀₀–1.1/CC < NiCo₂S₂₀₀–1.2/CC. Moreover, the detailed data can be found in Table S1, which are calculated based on our previous work [26]. For comparison, the CV curves of pure CC and pure NCS (co-NiCo₂S₄) are also supplied in Fig. S5. As shown, the CV curve (Fig. S5a) of CC displays as a straight line and the corresponding GCD profile (Fig. S5b) also behaves a straight line behavior, hinting its negligible influence on the capacitive properties of the above mentioned free-standing electrodes. But the co-NiCo₂S₄ exhibits analogous CV (Fig. S5a) and GCD (Fig. S5c) to the NiCo₂S₂₀₀-x/CC electrode, further suggesting the main phases of these electrodes are NiCo₂S₄ phase.

To evaluate the rate performance and reversibilities of the obtained electrodes, CV curves under various potential sweeping rates are also provided in Fig. 5b and Fig. S4. It is clear that all the CV curves exhibit stable contours when increasing the sweeping rate, with merely the peak current densities proportionally increasing, manifesting good rate performance of these electrodes. Detailly, the DSCs of these electrodes computed at various scan rates are tabulated in Table S1. On another aspect, when increasing the scan rate, the cathodic and anodic peaks move negatively and positively, respectively, with the redox peaks keeping nearly symmetric, hinting the good reversibilities of these electrodes.

Based on power law $i_p = av^b$, $b = 1$ and $b = 0.5$ separately denote the capacitive and diffusion-controlled charge storage processes [3]. In another word, if a diffusion-controlled process exists during the charge storage, when plotting the $i_p-v^{1/2}$ relationship, a straight line should be obtained. So, the $i_p-v^{1/2}$ plots of the NiCo₂S₂₀₀-x/CC electrodes are exhibited in Fig. 5f. It can be seen that all data in the cathodic and anodic directions comply with good linear relationships, confirming the diffusion-controlled kinetics of these electrodes. What is more, it is very

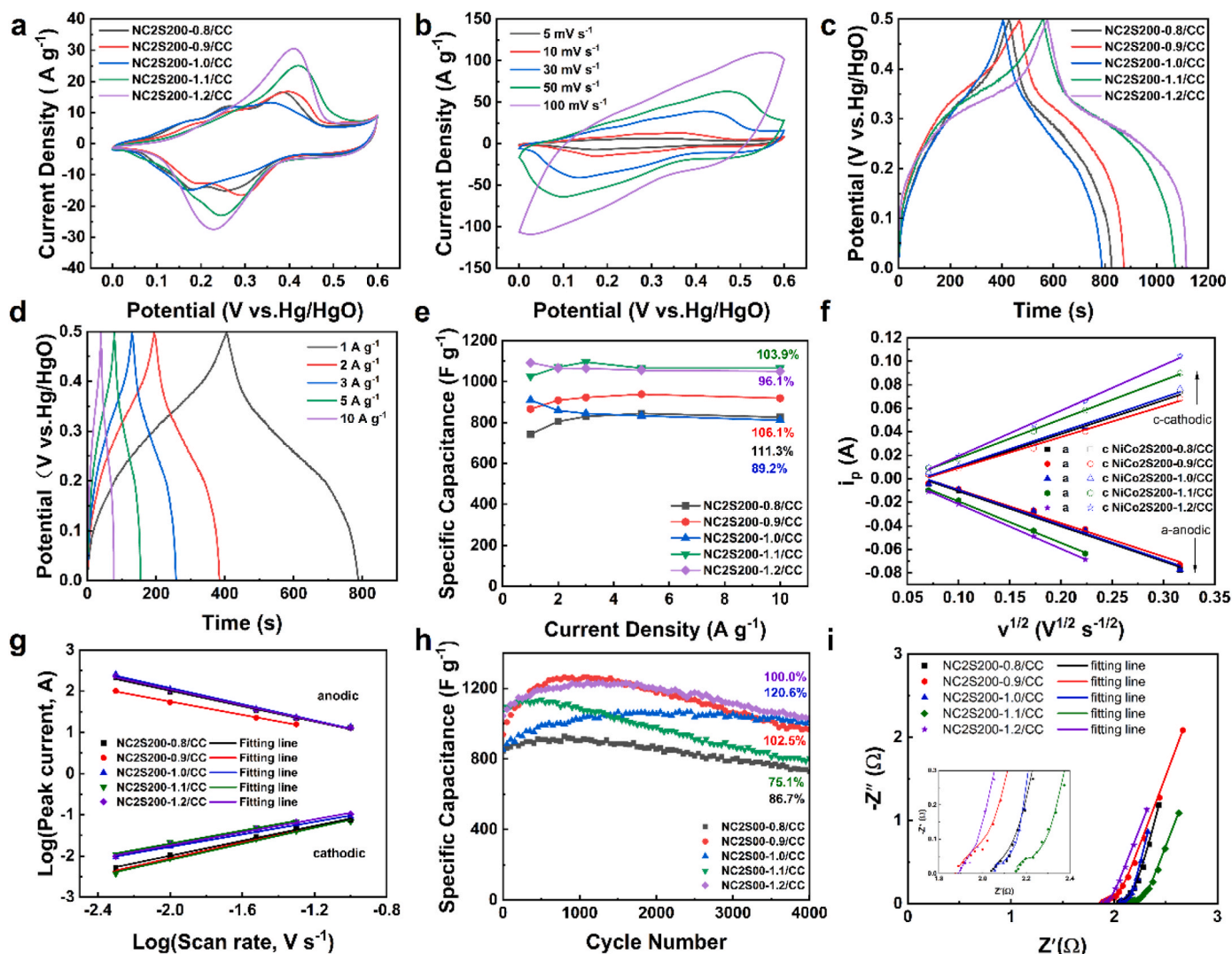


Fig. 5. (a) CV curves at 10 mV s⁻¹ within 0–0.6 V (vs.Hg/HgO), (c) GCD profiles at 1 A g⁻¹ in the potential window of 0–0.5 V (vs.Hg/HgO), (e) Rate performance, (f) i_p - $v^{1/2}$ plots, (g) $\log(i_p)$ - $\log(v)$ plots, (h) Cyclic stability curves and (i) Nyquist plots of the NiCo₂S_{200-x}/CC electrodes with $x = 0.8, 0.9, 1.0, 1.1$ and 1.2 , respectively. (b) CV curves at different scan rates from 5 to 100 mV s⁻¹ and (d) GCD profiles at different current densities from 1 to 10 A g⁻¹ for the NiCo₂S_{200-1.0}/CC electrodes.

important to distinguish the capacitive contribution from the total charge storage process in order to analyze the power properties of these electrodes [41]. Thus, by taking the logarithm of both sides of the power law, shown as equation (3), the value of b can be acquired by the slope of $\log(i_p)$ - $\log(v)$ line.

$$\log(i_p) = \log a + b \log(v) \quad (3)$$

Fig. 5g depicts the $\log(i_p)$ - $\log(v)$ relationships of the NiCo₂S_{200-x}/CC electrodes. As seen, all the electrodes display good linear relationships in the double logarithm plots. Additionally, both of the anodic and cathodic b values of these electrodes are supplementally listed in Table S2. It can be seen that the b values of the NiCo₂S_{200-0.9}/CC reach to 0.99 and 0.97 for the anodic and cathodic reactions, indicating its capacitive dominated charge storage mechanism that favors the electrode kinetics. In similar, the NiCo₂S_{200-0.8}/CC and NiCo₂S_{200-1.0}/CC electrodes also deliver b values ranging from 0.9 to 0.97. While, only 0.76–0.81 can be obtained for the NiCo₂S_{200-1.1}/CC and NiCo₂S_{200-1.2}/CC electrodes, implying the retard of charge storage kinetics which may be attributed to the enlarged grain clusters.

Fig. 5c depicts the GCD curves of the NiCo₂S_{200-x}/CC electrodes at a current density of 1 A g⁻¹. Apparently, there are two pairs of charge-

discharge plateaus in the GCD curves of the NiCo₂S_{200-0.8}/CC, NiCo₂S_{200-0.9}/CC and NiCo₂S_{200-1.0}/CC electrodes, respectively, in good accordance with the redox peaks in their CV curves. And the same scenarios can be observed for the other electrodes. However, these GCD curves appear quasi-triangular shapes which obviously differ from the battery-type behavior of nickel hydroxide [26,47], but resemble those of capacitive carbon materials [47,48], demonstrating the pseudocapacitive behaviors of NCSs prepared here. In addition, the GCD curves of the NiCo₂S_{200-1.0}/CC electrodes at various current densities are presented in Fig. 5d and the DSCs are computed according to equation (2) and detailedly listed in Table S3. It can be found that when increasing the current density from 1 to 10 A g⁻¹, the shapes of GCD curves keep nearly unchanged, demonstrating good rate performance of the electrode. Additionally, the changes of DSCs with the current densities of the NiCo₂S_{200-x}/CC electrodes can be found in Fig. 5e. As seen, when increasing the current density from 1 to 10 A g⁻¹, the capacitance retentions of 111.3 %, 106.1 %, 89.2 %, 103.9 % and 96.1 % can be assigned to the NiCo₂S_{200-x}/CC electrodes when $x = 0.8, 0.9, 1.0, 1.1$, and 1.2 , respectively, manifesting excellent rate performance of these electrodes. The exceptional increments of DSCs when elevating the current densities for the NiCo₂S_{200-0.8}/CC, NiCo₂S_{200-0.9}/CC and

NiCo₂S₂₀₀–1.1/CC electrodes should be attributed to the gradual activities of the active materials during the GCD tests. Theoretically, the good electronic conductivities and nanoscale porous morphologies of the deposited NCSs should be responsible for their excellent power capabilities. To test the limitation of rate capability of the NiCo₂S₂₀₀–1.0//electrode, the charge-discharges at 50 and 100 A g^{−1} were conducted and the GCD curves and DSCs are provided in Fig. S6. It can be seen that when increasing the current density to 50 A g^{−1}, serious IR drop can be observed, suggesting deteriorated electrode kinetics. When further charging-discharging at 100 A g^{−1}, the discharge plateau disappears, hinting the polarization further enlarges. In detail, the DSCs are 535.3 and 69.4 F g^{−1} at 50 and 100 A g^{−1}, respectively, manifesting that charging and discharging under ultrahigh current density is still great challenge of our electrode.

More importantly, the cyclic stabilities of the NiCo₂S₂₀₀-x/CC electrodes are measured under 5 A g^{−1} for 4000 cycles and the results are shown in Fig. 5h. As shown, when continuously charging and discharging for 4000 cycles, the DSC of the NiCo₂S₂₀₀–1.0/CC electrode increases from 852.6 F g^{−1} to 1028.2 F g^{−1}, corresponding to a retention rate of 120.6 %, transcending remarkably the other electrodes. In order to interpret the excellent cyclic performance, the SEM images of cycled NiCo₂S₂₀₀–1.0/CC electrode are supplied in Fig. S7. As seen, the porous nanosheet architecture has not been destroyed after charging-discharging cycling apart from local peeling-offs of active materials. Thus, it can be concluded that the robust structural stability should be a critical factor determining the cycling stability of binder-free electrode. Besides, the NiCo₂S₂₀₀–0.9/CC and NiCo₂S₂₀₀–1.2 electrodes also accomplish over 100 % capacitance retention, further verifying the good cycling stabilities of these electrodes. The above mentioned unusual augmentations of DSCs during the cycling tests may be related to the electrochemical activation process [49]. The above analyses illustrate that the electrochemical performance of electrodeposited NiCo₂S₂₀₀-x/CC electrodes are linked closely with the deposition potential because the microstructure and chemistry of the deposit evolve with the deposition potential. Excitingly, our electrochemical results can be comparable with some recently reported works about NCSs as tabulated in Table 1, exhibiting potential application prospects. To further verify the advantages of our free-standing electrodes compared to ex situ co-NiCo₂S₄ electrode, the cyclic performance of the co-NiCo₂S₄ electrode is also showed in Fig. S5d. As seen, when cycled at 3 A g^{−1} for 2000 cycles, the capacitance of the co-NiCo₂S₄ electrode can merely keep 38.1 %, which is far lower than our free-standing electrodes.

Fig. 5i exhibits the Nyquist plots of the NiCo₂S₂₀₀-x/CC electrodes

with the observed and fitting data concomitantly. Besides, the equivalent electric circuit for fitting the Nyquist plots is supplemented in Fig. S8. It is obvious that all the Nyquist plots consist of three sections, namely the high, middle and low frequency zones which are characterized by an intercept of horizontal axis, a compressed semicircle and a slash in from left to right. The circuit elements of these sections are depicted as R1, R2, CPE1 and W1, respectively, which represent the series impedance, charge transfer impedance, constant phase element and Warburg impedance [35]. Specifically, the values of fitted electric elements and Chi-squared values are listed in Table S4. As shown, the NiCo₂S₂₀₀–1.0/CC electrode possesses the smallest charge transfer impedance (R1 = 0.0229 Ω), indicating excellent electronic transportation kinetics which should be benefited from its porous nanosheet-like topography and high conductivity. It is worth noting that the R1 values of the other electrodes are also very small, indicating their competitive topographies and conductivities. However, from the view point of cyclic stability, the NiCo₂S₂₀₀–1.0/CC electrode exhibits more glamour.

In order to investigate the potential application of the NiCo₂S₂₀₀–1.0/CC electrode, a HC consists of the NiCo₂S₂₀₀–1.0/CC and AC electrodes (in a mass ratio of 0.81: 2.01) are assembled and tested. Fig. 6a exhibits the CV curves of the HC in different potential windows (PWs) at 10 mV s^{−1}. As shown, when enlarge the PW from 1.0 to 1.6 V, no obvious oxygen evolution reaction (OER) can be detected, indicating the reasonability of a PW of 1.6 V for the HC. As seen in Fig. 6b, when altering the scan rate from 5 to 100 mV s^{−1}, the redox current density increases proportionally meanwhile keep quasi-rectangular shape, suggesting good pseudocapacitive rate capability of the HC. The initial GCD profiles under various current densities (as shown in Fig. 6c) present inverse quasi-triangular shapes further reveal the pseudocapacitive features of the HC. The charge plateau appeared in the GCD curve of 1 A g^{−1} is relevant to the OER sidereaction which can be suppressed by elevating the current density. The dependence of DSC on current density is presented in Fig. 6d, details, the DSCs of the HC are 73.8, 65.8, 55.3 and 45.1 F g^{−1} at 1, 2, 4 and 8 A g^{−1}, respectively. As shown in Fig. 6e, when charging-discharging at 2 A g^{−1} for 10000 cycles, the DSC can be retained 75.5 %, manifesting good cyclic performance. Moreover, this result is superior to some previously reported NCSs based HCs, as exhibited in Table S5. Furthermore, the Ragone plot of the HC demonstrates a high energy density of 26.2 Wh kg^{−1} at a power density of 821.1 W kg^{−1}, which can maintain 16.0 Wh kg^{−1} when the power density increase to 5767.2 W kg^{−1}, indicating good power capability of the device. It can be seen that the NiCo₂S₂₀₀–1.0/CC//AC device can deliver comparable energy density and power density when compared with some reported HCs, such as NiCo₂S₄/RGO//AC (24.4 Wh kg^{−1} at 750 W kg^{−1}) [53], NiCo₂S₄@MnS//AC (23.3 Wh kg^{−1} at 725 W kg^{−1}) [54], NCS/rGO/CNT//AC (21.1 Wh kg^{−1} at 726 W kg^{−1}) [49], NiCo₂S₄/d-Ti₃C₂//AC (22.6 Wh kg^{−1} at 400 W kg^{−1}) [55], NiCo₂Se₄/MXene//AC (22.4 Wh kg^{−1} at 800 W kg^{−1}) [56], SAC@NiCo_{0.95}Mn_{0.05}-LDH//AC (23.7 Wh kg^{−1} at 325 W kg^{−1}) [57] and CuO@S-NiV LDH//NF@AC (26.0 Wh kg^{−1} at 1001 W kg^{−1}) [58]. Therefore, the NiCo₂S₂₀₀–1.0/CC electrode can be an eligible free-standing electrode for supercapacitors.

4. Conclusions

Herein, a series of NCSs with various chemical compositions and morphologies have been immediately electrodeposited on CC substrate within electrolyte solutions containing nickel cobalt sulfates and thiourea only by tuning the deposition potential. The electrode prepared under −1.0 V (vs. Ag/Ag/Cl) exhibited ordered nanosheet arrays which facilitated the mass/electron transportation, thus delivered excellent electrochemical performance. In detail, when tested as a supercapacitor electrode under 1 A g^{−1}, a DSC of 909.5 F g^{−1} can be achieved and can maintain 811.3 F g^{−1} at 10 A g^{−1}. When charging and discharging at 5 A g^{−1} for 4000 cycles, the DSC can retain 120.6 %, exhibiting unique

Table 1

Electrochemical performance comparisons of some NCS based electrodes reported previously.

Electrodes	Synthesis	Rate performance		Cycling performance	Ref.
		1 A g ^{−1} (F g ^{−1})	10 A g ^{−1} (F g ^{−1})		
NiCo ₂ S ₄ /NF	Colloidal	1790.8	1408.0	10 A g ^{−1} , 2000, 86.8 %	[50]
NiCo ₂ S ₄	ST	1386.0	845.0	10 A g ^{−1} , 5000, 80.1 %	[51]
NiCo ₂ S ₄ -2D Carbyne/NF	ST	2507.2	1897.0	No	[52]
CoNi ₂ S ₄	HT	1980.0	1569.0	10 A g ^{−1} , 5000, 100.0 %	[4]
CNS/PPy/CP	UPED	2004.0	No	5 A g ^{−1} , 1000, 90.0 %	[29]
NiCo ₂ S ₄ /CP	ED	1410.0	1300.0	5 A g ^{−1} , 2000, 98.0 %	[30]
NCS/CC	ED	909.5	811.3	5 A g ^{−1} , 4000, 120.6 %	This work

Notes: ST solvothermal, HT hydrothermal, UPED unipolar pulse electrodeposition, ED electrodeposition, NF nickel foam, CP carbon paper, CC carbon cloth.

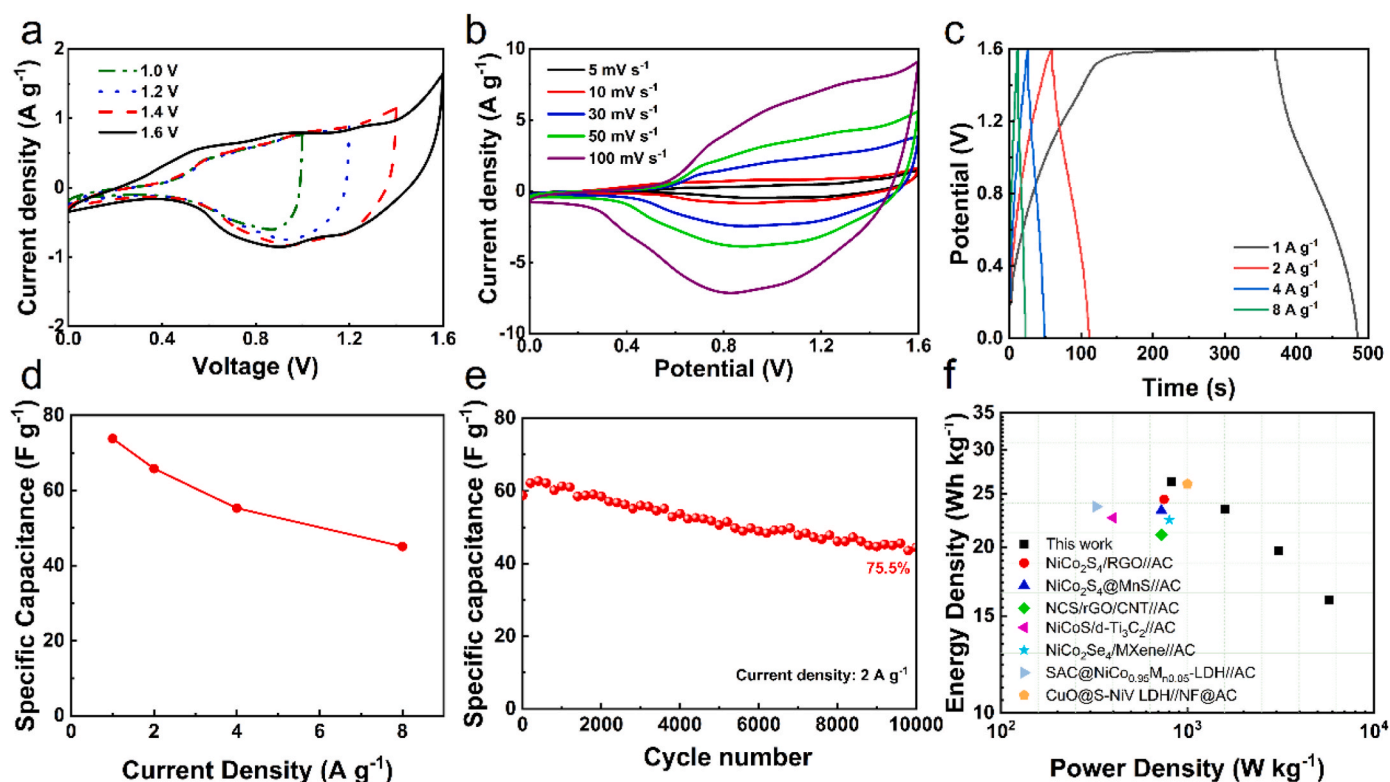


Fig. 6. (a) CV curves at 10 mV s^{-1} within different potential windows from 1.0 to 1.6 V, (b) CV curves at various scan rates from 5 to 100 mV s^{-1} within 0–1.6 V, (c) Initial GCD profiles under different current densities from 1 to 8 A g^{-1} in the potential window of 1.6 V, (d) Rate performance, (e) Cyclic stability and (f) Ragone plots of the $\text{NiCo}_2\text{S}_{200}-1.0/\text{CC}/\text{AC}$ HC.

cycling capability. The $\text{NiCo}_2\text{S}_{200}-1.0/\text{CC}/\text{AC}$ device also presents good cycling capability (75.5 % retention at 2 A g^{-1} for 10000 cycles), high energy density (26.2 Wh kg^{-1} , 821.1 W kg^{-1}) and high power density (5767.2 W kg^{-1}), demonstrating potential application prospects. This work provides a new strategy for constructing high performance free-standing electrodes for supercapacitors.

CRediT authorship contribution statement

You Wang: Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Lianzhen Liao:** Data curation, Software, Writing – original draft. **Quanwen Xie:** Visualization, Investigation, Methodology, Resources. **Zhoulan Yin:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.solidstatesciences.2025.107941>.

Data availability

Data will be made available on request.

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